

One Lyapunov Ratio, Three Global Phases: The May–Leonard Coexistence–Cycle Atlas

HCS-C358 source-local reconstruction

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Abstract

For the three-species cyclic May–Leonard competition flow in the strict intransitive chamber, we prove an all-interior-orbit classification. The normalized product $\mathcal{R} = xyz/(x+y+z)^3$ has a derivative whose sign is exactly that of $2 - a - b$. Below the critical plane it yields global convergence to coexistence. On the plane it combines with an exact logistic time change to give a foliation by periodic rock–paper–scissors leaves and an elliptic period quadrature. Above the plane, the positive diagonal is the complete stable set of the unstable coexistence point, while every other interior orbit has the full oriented boundary heteroclinic cycle as its omega-limit set and has diverging residence times. Exact rational and symbolic receipts audit the identities and conventions without replacing the global proof. No literature-priority or arithmetic-target claim is made.

Revision certificate. Round-two heteroclinic exhaustion, evidence, and Route-A closure.

1 Model, chamber, and main result

For $a, b \geq 0$, consider

$$\begin{aligned}\dot{x} &= x(1 - x - ay - bz), \\ \dot{y} &= y(1 - bx - y - az), \\ \dot{z} &= z(1 - ax - by - z).\end{aligned}\tag{1}$$

We work in the strict cyclic chamber

$$(a - 1)(b - 1) < 0.\tag{2}$$

Put $S = x + y + z$, $Q = xy + yz + zx$, and, in the interior, $\mathcal{R} = xyz/S^3$. Let

$$p = \frac{1}{1+a+b}(1, 1, 1), \quad E_1 = (1, 0, 0), \quad E_2 = (0, 1, 0), \quad E_3 = (0, 0, 1).$$

Theorem 1 (Complete cyclic-chamber atlas). *The closed positive octant is forward invariant; its interior is preserved; and every positive semiorbit is bounded and forward complete. The point p is the unique interior equilibrium. For every interior initial state:*

- (i) *If $a + b < 2$, the solution converges to p .*
- (ii) *If $a + b = 2$, then S obeys a logistic equation. With $(u, v, w) = (x, y, z)/S$, $\delta = 1 - a = b - 1$, and $d\tau = S dt$, the normalized flow is*

$$u_\tau = \delta u(v - w), \quad v_\tau = \delta v(w - u), \quad w_\tau = \delta w(u - v).\tag{3}$$

Its centre is fixed, and each level $uvw = h$, $0 < h < 1/27$, is one periodic orbit. If $r_-(h) < 1/3 < r_+(h)$ are the roots in $(0, 1)$ of $r(1 - r)^2 = 4h$, its least τ -period is

$$T(h) = \frac{2}{|\delta|} \int_{r_-(h)}^{r_+(h)} \frac{dr}{\sqrt{r\{r(1 - r)^2 - 4h\}}}.\tag{4}$$

Every original noncentral solution approaches a unique phase translate of its normalized periodic orbit.

- (iii) *If $a + b > 2$, the complete interior stable set of p is the positive diagonal. Every other interior orbit has as its omega-limit set the full three-connection boundary heteroclinic cycle and spends unbounded successive times near its axial saddles. For $a < 1 < b$ its orientation is $E_1 \rightarrow E_3 \rightarrow E_2 \rightarrow E_1$; for $b < 1 < a$ the arrows reverse.*

At the closure point $a = b = 1$, all ratios are fixed and the simplex $S = 1$ consists entirely of equilibria. The separate walls $a = 1$ or $b = 1$ are nonhyperbolic and are not included in (2).

2 Positive flow and the source identity

Each coordinate in (1) is itself times a locally Lipschitz function. Thus a zero coordinate remains zero, whereas a positive one cannot vanish at finite time. Since $a, b \geq 0$,

$$\dot{x} \leq x(1-x), \quad \dot{y} \leq y(1-y), \quad \dot{z} \leq z(1-z).$$

Scalar comparison proves boundedness and continuation for all positive time. For some $C = C(a, b) > 0$, summing (1) also gives $\dot{S} \geq S - CS^2$. Consequently an interior semiorbit eventually lies in a compact annulus bounded away from the origin.

At an interior equilibrium,

$$\begin{pmatrix} 1 & a & b \\ b & 1 & a \\ a & b & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The determinant is

$$(1+a+b)(1+a^2+b^2-a-b-ab),$$

and twice the second factor is $(a-b)^2 + (a-1)^2 + (b-1)^2$. It vanishes only at $a = b = 1$. Subtracting cyclic rows therefore gives the sole positive solution p in the strict chamber.

Lemma 1 (Normalized-product identity). *Along every interior solution,*

$$\dot{S} = S - S^2 + (2-a-b)Q, \tag{5}$$

$$\frac{d}{dt} \log(xyz) = 3 - (1+a+b)S, \tag{6}$$

$$\frac{d}{dt} \log \mathcal{R} = (2-a-b) \frac{S^2 - 3Q}{S}, \tag{7}$$

$$S^2 - 3Q = \frac{1}{2} \{(x-y)^2 + (y-z)^2 + (z-x)^2\}. \tag{8}$$

Proof. Adding (1) and using $x^2 + y^2 + z^2 = S^2 - 2Q$ gives (5). Dividing each equation by its coordinate and adding gives (6). Equation (7) is (6) minus $3\dot{S}/S$, and direct expansion gives (8). $\square$

The derivative in (7) is strict away from the positive diagonal whenever $a+b \neq 2$. In particular there can be no nonconstant interior periodic orbit off the critical plane.

3 Subcritical global coexistence

Assume $a+b < 2$. By Lemma 1, $\mathcal{R}$ increases and is bounded above by $1/27$. The bound $\mathcal{R}(t) \geq \mathcal{R}(0) > 0$ keeps $(x/S, y/S, z/S)$ away from the simplex boundary, while the annulus estimate controls S . Hence the orbit is eventually contained in a compact subset of the interior. LaSalle's invariance principle applied to $\log \mathcal{R}$ places its omega-limit set on $x = y = z$. That diagonal is invariant, and $x = y = z = r$ reduces (1) to

$$\dot{r} = r\{1 - (1+a+b)r\}. \tag{9}$$

The only compact complete diagonal orbit in the positive annulus is the equilibrium $r = (1+a+b)^{-1}$. Thus the omega-limit set is $\{p\}$ and the whole orbit converges to p , proving Theorem 1(i).

4 Critical normalization and the elliptic period

Now let $a+b = 2$. Equation (5) becomes $\dot{S} = S(1-S)$, so, with $S_0 = S(0)$,

$$S(t) = \frac{S_0 e^t}{1 + S_0(e^t - 1)}, \quad \tau(t) = \log\{1 + S_0(e^t - 1)\}. \tag{10}$$

Direct differentiation gives $\dot{\tau} = S$ and $\tau(t) - t \rightarrow \log S_0$.

Differentiating $u = x/S$, cyclically, proves (3). The components of (3) sum to zero and $d(uvw)/d\tau = 0$. On the open simplex, the sole equilibrium is the centre. For $0 < h < 1/27$, the regular set

$$\Gamma_h = \{u + v + w = 1 : u, v, w > 0, uvw = h\}$$

is a compact circle surrounding the centre. Its vector field never vanishes, so it is exactly one periodic orbit.

To compute the period, use $v + w = 1 - u$ and $vw = h/u$:

$$(u_\tau)^2 = \delta^2 u^2 (v-w)^2 = \delta^2 u \{u(1-u)^2 - 4h\}. \tag{11}$$

The function $r(1-r)^2$ rises from zero to $4/27$ on $(0, 1/3)$ and falls back to zero on $(1/3, 1)$. Its two roots at height $4h$ are precisely the turning values r_-, r_+ . Integrating (11) from one turning point to the other and back yields (4); this is the least period because the vector field traverses Γ_h once.

Let $U(\tau)$ be the periodic normalized solution with $U(0) = (u(0), v(0), w(0))$. Equation (10) gives $(x, y, z)(t) = S(t)U(\tau(t))$. Since $S(t) \rightarrow 1$ and $\tau(t) - (t + \log S_0) \rightarrow 0$, uniform continuity of U gives

$$(x, y, z)(t) - U(t + \log S_0) \rightarrow 0. \quad (12)$$

This proves the stated asymptotic phase, uniquely modulo the least period.

5 The diagonal exception above criticality

Let $a + b > 2$. Linearization at p has the radial eigenvalue -1 and a complex tangent pair satisfying

$$\operatorname{Re} \lambda = \frac{a + b - 2}{2(1 + a + b)} > 0, \quad (\operatorname{Im} \lambda)^2 = \frac{3(a - b)^2}{4(1 + a + b)^2}. \quad (13)$$

Thus the local stable manifold is one-dimensional. The positive diagonal is invariant and tangent to the stable eigenspace. Uniqueness in the stable manifold theorem identifies them locally. If an interior orbit converges to p , it eventually belongs to this local stable manifold; uniqueness of the flow then puts its entire orbit on the diagonal. Conversely, (9) shows that every positive diagonal orbit converges to p . Hence the complete interior stable set is exactly the positive diagonal.

For every other interior orbit, (7) makes $\mathcal{R}$ strictly decreasing. If its limit were positive, the normalized coordinates would remain away from the boundary. LaSalle's principle applied to $-\log \mathcal{R}$ would then force the omega-limit set to p , contradicting the stable-set classification. Thus

$$\mathcal{R}(t) \rightarrow 0. \quad (14)$$

Since S stays in a positive compact annulus, the omega-limit set is a nonempty compact invariant subset of the nonzero boundary.

6 Boundary exhaustion and residence times

Suppose first that $a < 1 < b$. On $z = 0$,

$$\frac{d}{dt} \log \frac{y}{x} = -(b - 1)x - (1 - a)y < 0. \quad (15)$$

There is no positive equilibrium in this coordinate plane. The Dulac multiplier $(xy)^{-1}$ has divergence $-1/x - 1/y < 0$ there, so there is no positive planar cycle. By Poincaré–Bendixson, the absence of a positive planar equilibrium, and the strict ratio in (15), the only nonconstant bounded complete planar orbit whose closure avoids the origin is the positive branch of the unstable manifold of E_2 , connecting E_2 to E_1 . Cyclic permutation gives the three connections

$$E_1 \rightarrow E_3 \rightarrow E_2 \rightarrow E_1.$$

Every other positive-plane orbit either has the origin in its backward closure or escapes compact sets in negative time. A non-equilibrium axial orbit also has the origin in its backward closure.

Let Ω be the omega-limit set of the interior orbit. Standard flow properties make Ω compact, connected, invariant, and internally chain transitive. It is bounded away from the origin. The preceding planar classification and (14) therefore confine Ω to the three axial equilibria and the three connections (16). It cannot be one axial point: at each E_i the stable manifold in the closed octant is the incoming coordinate face, whereas the omitted-species eigenvalue is $1 - a > 0$. Nor can it be a proper closed directed subchain. Choose disjoint small neighbourhoods of the axial points and compact middle arcs on the included connections. On each middle arc the derivative in (15), or its cyclic analogue, has a uniform strict sign, and the flow crosses the two boundary sections only in the directed order. For sufficiently small jump size, no long-time pseudo-orbit within a proper subchain can return from its terminal axial neighbourhood to its initial one. Internal chain transitivity leaves exactly the full cycle (16). This proves Ω equals that cycle, not merely that its distance from the boundary tends to zero.

The orbit enters arbitrarily small fixed neighbourhoods of every E_i . At successive entries its outgoing coordinate tends to zero. In such a neighbourhood that coordinate grows no faster than $1 - a + \varepsilon$. The time needed to reach a fixed exit section is thus at least a constant multiple of the logarithm of the reciprocal entry coordinate, and tends to infinity. Swapping y and z swaps a, b and reverses (16), completing Theorem 1(iii).

7 Closure faces, evidence, and claim boundary

On each coordinate axis, (1) is the scalar logistic flow. The origin is repelling relative to the positive octant. At $a = b = 1$, all equations have the form $\dot{x}_i = x_i(1 - S)$: ratios are constant, S is logistic, and the whole simplex $S = 1$ is equilibria. On either separate wall $a = 1$ or $b = 1$, an axial transverse eigenvalue vanishes; those dominance walls are recorded but excluded from the strict theorem.

The exact certificate contains 12 parameter rows, 72 invariant rows, 24 critical rows, 22 logistic rows, six orientation rows, and five boundary rows. An independent implementation reconstructs all rows and strict serialization contracts; a separate symbolic derivation checks 19 identities; two isolated replays agree byte for byte; and 67 repaired-hash mutations are rejected. These are receipts for algebra and convention. They do not prove LaSalle exhaustion, stable-manifold uniqueness, periodic foliation, or the omega-limit theorem; the preceding arguments do.

May and Leonard supply the original cyclic-competition lineage [1]; Blé, Castellanos, Llibre, and Quilantán provide later integrability and global dynamics context [2]. We make no priority claim.

The strict Route-A tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).$$

The source has a continuum of critical periodic leaves but no intrinsic rational-prime objects, prime-power repetition law, logarithmic-prime clock, target determinant, or natural target-zero quantization. Route A is rejected and Route B is locked. In particular, no target arithmetic local data, Euler factors, bad-prime data, root number, automorphy, target divisor or counting law, target functional equation, target-zero match, or Hilbert–Pólya operator is claimed.

References

- [1] R. M. May and W. J. Leonard, “Nonlinear aspects of competition between three species,” *SIAM J. Appl. Math.* 29 (1975), 243–253. doi:10.1137/0129022.
- [2] G. Blé, V. Castellanos, J. Llibre, and I. Quilantán, “Integrability and global dynamics of the May–Leonard model,” *Nonlinear Anal. Real World Appl.* 14 (2013), 280–293. doi:10.1016/j.nonrwa.2012.06.004.